

Assignment: Linux

Module: 1 - Linux server - Understand and use essential tools

1. What is the minimum number of partitions you need to install Linux?

Ans : For a basic Linux installation, minimum of 1 partition

2. Explain About Chmod Command.

Ans : The chmod command in Linux is used to change file and directory permissions.

chmod stands for change mode.

It controls who can:

- Read (r)
- Write (w)
- Execute (x)

3. How to check Linux memory utilization.

Ans : To check memory utilization in Linux there are several commands.

- free -h
- top
- htop
- cat /proc/meminfo
- vmstat

4. Use grep to search for specific patterns in files.

Ans : The grep command is a powerful Linux utility used to:

- Search text patterns
- Filter output
- Analyze logs
- Work with regular expressions

5. Get Connecting on a linux server by ssh.

Ans :

- Open Terminal
- Type SSH command:
`ssh username@server_ip`
- Press Enter
- Enter password
- Login successful

6. Create 5 files in the /tmp directory, and then use tar and gzip to bundle and compress the files.

Ans :

- `touch file1 file2 file3 file4 file5`
- `tar -cvf files.tar file1 file2 file3 file4 file5`
- `gzip files.tar`
- `files.tar.gz`

7. Describe the root account.

Ans : The root account in Linux is the administrator account with full control over the system.

- Root can modify or delete any file.
- Wrong commands may damage the system.
- Used mainly for system administration tasks.

8. What is shell?

Ans : A shell is a command-line interface in Linux that allows users to communicate with the operating system.

9. What is Linux?

Ans : Linux is an open-source operating system (OS) based on Unix.

It manages hardware and software resources and allows users to interact with the computer using commands or graphical interface.

10. What is Bash?

Ans : Bash (Bourne Again Shell) is a Linux command-line shell and scripting language.

It is used to run commands, execute programs, and write scripts in Linux and Unix systems.

11. You have a new empty hard drive that you will use for Linux. What is the first step you use.

Ans : The first step is to partition the hard drive before installing Linux.

12. Write the Linux command to show the current working directory.

Ans : `pwd` stands for Print Working Directory.

13. Write the Linux command to get help with various options.

Ans : `man` stands for manual.

It shows detailed help, options, and usage of a command.

14. Write the linux comman! to display what all users are currently doing.

Ans : `w` shows logged-in users and what they are currently doing.

15. write the Linux command to get information about the operating system.

Ans : `uname` shows system information.

16. Write the Linux command to create a hard link of a file.

Ans : `ln` → command to create links.

17. Write the Linux command to create a soft link of a file as well as Directory.

Ans : `ln -s`

18. Write the Linux command! to search for specific pattern in a file.

Ans : grep

19. Write the Linux command to show the use of basic regular expressions using grep command.

Ans :

- ^ → start of line
- \$ → end of line
- . → single character
- → repeated characters
- [] → set of characters